

Support-Aware Review Queues for Crop–Weed Vision: A Multitemporal Single-Field Case Study

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Abstract

Finite crop–weed review queues can lose plants during candidate formation, role qualification, ranking, or budget allocation. We evaluated a support-aware queue framework on the official WeedsGalore spatial split (104/26/26 RGB tiles) from one maize field; the test set contains two held-out spatial patches observed across four acquisition dates. Target-training masks supported proposal selection, candidate labels, and a semantic baseline, while validation selected model settings. Proposal commissioning increased spatial weed recall from 0.2973 to 0.5713 and role-qualified recall from 0.1928 to 0.3026, with candidate burden rising from 30.85 to 90.42 per tile. A component-masked DINOv2 ranker achieved candidate-level role-qualified AUC 0.9665 and average precision 0.8813. Across five H200 training seeds, target-semantic fixed queues achieved precision 0.7168 ± 0.0260 and one-to-one recall 0.2027 ± 0.0057 . An exploratory 5–50 candidates/tile allocation increased these values to 0.8939 ± 0.0249 and 0.2578 ± 0.0090 , but reduced mean worst-date recall from 0.1100 to 0.0473. Date-balanced allocation provided an intermediate operating point: precision 0.7710, one-to-one recall 0.2192, and worst-date recall 0.1163. Its patch-stratified recall effect was negative in one capture-index

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group and positive in the other. The resulting single-field evidence shows that proposal support and allocation constraints jointly determine queue utility, and that pooled gains require explicit temporal and spatial-group accounting.

Keywords: weed detection, spatial validation, candidate proposal, self-supervised representation, review queue, precision agriculture

1. Introduction

Site-specific weed management links sensing, plant localization, crop–weed discrimination, and intervention [1, 2]. A classifier score captures one part of this operational chain. Object-proposal evaluation relates recall to proposal count and overlap [3], whereas precision-at-the-top learning optimizes performance within a constrained ranking budget [4]. Selective prediction formalizes the complementary trade-off between coverage and error [5, 6], and learning-to-defer methods formalize assignment between automated and human decision makers [7, 8]. A crop–weed review queue may lose an instance during candidate formation, role qualification, rank ordering, or budget truncation. Separating these stages connects candidate-level discrimination to end-to-end instance recovery without treating a queue entry as an agronomic decision.

Agricultural images contain strong spatial and temporal dependence. Random image partitions can overstate generalization when neighboring observations share soil, illumination, management, or canopy structure [9]. Public crop–weed datasets also vary in crop, sensor, viewpoint, acquisition date, image scale, and annotation contract [10, 11, 12, 13, 14]. Cross-crop semantic transfer [15], sequence-based localization [16], and field-focused reviews [17] emphasize the complete sensing and decision chain. We use the official WeedsGalore spatial split to evaluate repeated acquisitions across spatial patches within one field.

Frozen self-supervised features offer a practical representation layer when target-field annotation is limited. DINOv2 provides transferable visual features from large-scale self-supervised pretraining [18]. Agricultural active learning links selective annotation to redundancy and class imbalance [19], while uncertainty-aware segmentation motivates routing ambiguous regions to review [20]. Recent weed-vision research reduces supervision through point-labelled instance masks, semi-supervised learning with unlabelled maize imagery, and one-shot segmentation [21, 22, 23]. These methods quantify

how annotation design changes segmentation accuracy and labour. A review queue answers a different operational question: after a model proposes image regions, which weed instances remain available after role qualification, finite-budget ranking, and cross-tile allocation? This stage-wise accounting complements segmentation and annotation-efficiency results by localizing losses that aggregate pixel or instance metrics do not identify.

Confidence ordering can change under covariate shift, so a globally ranked batch requires explicit checks of score comparability across acquisition groups [24, 25, 26]. Constrained ranking provides an adjacent formal perspective on balancing score utility and group exposure [27]. Multispectral WeedsGalore results demonstrate additional information from red-edge and near-infrared bands [28]. Together, these studies motivate an RGB queue analysis that measures where proposal support, representation, allocation, and review shape recoverable instances.

We model crop and weed as scene-dependent agricultural roles. Taxonomic identity (family, genus, and species) and management morphotype (grass-like, broadleaf, sedge, and other) remain separate output axes. The present experiments quantify role-specific candidate recovery.

We introduce a support-aware queue framework with three measurable layers: spatial proposal recall, role-qualified proposal recall, and queued recall after ranking and allocation. Shared-ranker-weight comparisons isolate candidate formation, and all-candidate fitting aligns ranker supervision with deployed support. A target-semantic baseline with equal target-mask access measures direct target-domain component formation. Fixed and quota-constrained allocations are evaluated at matched review budgets, with validation-selected, date-balanced, worst-date, and spatial-block diagnostics. Many-to-one and one-to-one recovery, IoU, crop exposure, simulated label budgets, and per-tile heterogeneity complete the evidence chain.

2. Materials and methods

2.1. Data lineage, sensor scope, and spatial split

The source specialist used the public SugarBeets2016 RGB images and masks [11]. The target evaluation used 156 annotated 600×600 pixel tiles from the public WeedsGalore release [14]. WeedsGalore contains four acquisitions from one maize field and 48 recorded locations: 36 locations were observed on the first three dates and 12 on all four dates. Its official 8/2/2 spatial-patch

allocation yields 104 training, 26 validation, and 26 test tiles. The test split contains 1,712 weed and 256 crop instances in the released raster masks.

All inference used RGB to match the SugarBeets source specialist and isolate cross-dataset transfer under a common sensor contract. Red-edge and near-infrared channels define a complementary multispectral extension.

Raster enumeration produced 2,168 crop and 10,029 weed instances across the 156 tiles, whereas the dataset paper reports 2,169 and 10,031. Direct mask checks found zero semantic-instance inconsistencies and zero excluded instance identifiers. All analyses use the reproducible raster denominators, with the evaluated archive hash reported in the supplement.

Figure 1 defines the information flow. Target RGB enters proposal generation, official-training dense masks support proposal selection and candidate-label derivation, and validation selects regularization, epoch, semantic threshold, minimum component area, and the transferable DINOv2 queue threshold. Every queue was formed from images, scores, and tile identities before its offline scorer accessed test labels.

Table 1 separates the analysis stages by chronology. The core proposal, ranking, matching, and fixed- K configuration was archived before its first official-test score. Target-semantic training, repeated-seed evaluation, quota sensitivity, exact component-mask replay, and group-stratified allocation were specified after the core test had been accessed and are reported as exploratory analyses. The archive contains timestamped configurations and checksums rather than a public immutable preregistration. None of the exploratory results altered the core proposal grid, matching contract, feature definitions, or fixed- K results.

2.2. Source RGB specialists and proposal settings

The foreground and crop/weed semantic specialists were ResNet-18 U-Net-style networks initialized with torchvision ImageNet-1K V1 weights [29]. RGB was scaled to $[0, 1]$ and normalized by the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). Both models used the 7,050-image source training partition. The foreground model used 32,768 deterministic 128×128 patches, binary cross-entropy with positive weight 30 plus soft Dice, and 10 epochs. The semantic model used 16,384 role-balanced patches, class-weighted cross-entropy with weights (0.2, 1, 1) plus crop/weed Dice, and 8 epochs. Training used deterministic patches without stochastic augmentation. AdamW (learning rate 8×10^{-4} , weight decay 10^{-4}), bfloat16 autocasting, fixed seeds, gradient clipping at 5, and final-epoch checkpoints

Table 1: Analysis chronology. Times use China Standard Time (UTC+8). “First test output” denotes the earliest archived official-test result for the corresponding analysis family.

Analysis family	Archived specification	First test output	Interpretation
Proposal, ranker, matching, fixed K	8 Aug 2026, 22:02	10 Aug 2026, 14:03	Recorded core analysis
Target-semantic reference	10 Aug 2026, 17:16	10 Aug 2026, 20:09	Post-test exploratory
Five-seed H200 repetitions	12 Aug 2026, 15:44	12 Aug 2026	Post-test exploratory
Quota grid, exact replay, group audits	12 Aug 2026	12 Aug 2026	Post-test exploratory

completed both runs. The 901-image source validation partition subsequently provided operating-point diagnostics. The foreground diagnostic reached maximum pixel IoU 0.3821 at threshold 0.99 (precision 0.5788, recall 0.5293). At threshold 0.70, recall was 0.9631 and precision was 0.0479. The semantic diagnostic reached three-class macro IoU 0.4425. Its role threshold yielded balanced accuracy 0.7224, crop recall 0.5524, and weed recall 0.8924. These values characterize source-domain pixels; target queue metrics appear in Section 3.

For a target RGB tile, the frozen specialists produced foreground probability $f(u)$ and semantic probabilities $p_c(u)$ and $p_w(u)$. A proposal foreground mask at threshold T was decomposed into connected components, and components smaller than area A were removed. Each setting is therefore defined by the pair (T, A) . Candidate features never read target masks.

The pre-existing external-inference configuration used $T = 0.95$ and $A = 32$ pixels and was fixed before the official spatial reanalysis. Commissioning evaluated a 12-setting grid on the 104 official-training tiles. Feasible settings satisfied a training-set mean burden of at most 100 candidates/image, while per-image counts remained unrestricted. The objective maximized training role-qualified weed proposal recall. Ties favored lower mean burden, higher threshold, and larger area, in that order. This procedure selected $T = 0.70$ and $A = 32$ pixels, with 95.07 candidates/image, spatial recall 0.5271, and role-qualified recall 0.2856 on training. The 30 May training burden reached 130.38/image. On official test, the median was 89, the interquartile range was 60–114.25, and the maximum was 183. Training masks exclusively determined

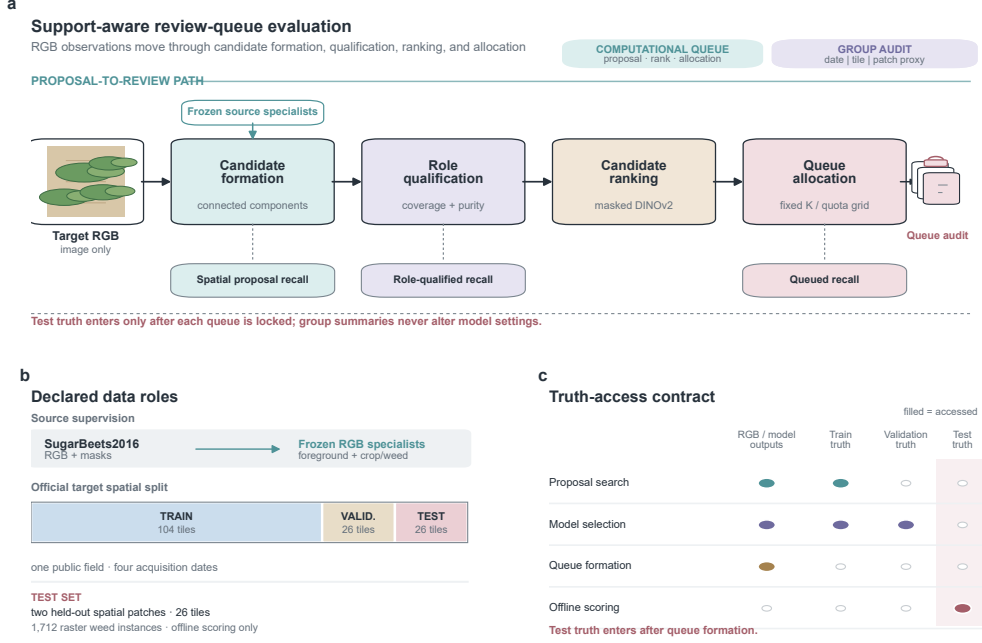


Figure 1: Information architecture of the support-aware queue analysis. (a) The RGB-only path separates candidate formation, role qualification, ranking, allocation, and group-stratified queue auditing. (b) Source supervision and the official WeedsGalore 104/26/26 spatial roles; the 26-tile test set comprises two held-out spatial patches observed across four acquisition dates. (c) Truth-access contract. Training truth supports proposal search and model selection, validation truth is restricted to setting selection, queue formation uses only RGB and model outputs, and test truth is accessed after a queue is locked for offline scoring.

proposal-setting selection.

2.3. Candidate contract and non-interchangeable recall layers

Let I_j be a truth instance and C_i a proposal. Spatial coverage is

$$\text{cov}(I_j, C_i) = \frac{|I_j \cap C_i|}{|I_j|}. \quad (1)$$

Spatial proposal recall counts I_j when any candidate has $\text{cov} \geq 0.50$, irrespective of candidate role. Role-qualified recall additionally requires at least 0.50 labeled-pixel coverage and 0.90 same-role purity in the same candidate.

Queued recall applies the corresponding contract to selected candidates. We report the three layers separately to identify support, qualification, and allocation effects. The 0.50/0.50/0.90 values are fixed engineering thresholds for offline candidate accounting, not agronomic treatment or actuation thresholds; the full sensitivity grid below shows how conclusions vary when they change.

Many-to-one recall permits one candidate to cover multiple instances. We additionally solve maximum-cardinality one-to-one candidate–instance matching and report standard intersection-over-union sensitivity. The full contract grid crosses instance coverage $\{0.25, 0.50, 0.75\}$, labeled coverage $\{0.25, 0.50, 0.75\}$, and same-role purity $\{0.50, 0.75, 0.90\}$. Split and merge summaries at overlap thresholds 0.10, 0.25, and 0.50 quantify contract sensitivity.

2.4. *Rankers and support-aware targets*

All linear models were fitted with standardized features and logistic loss. The discrimination comparison selected $C \in \{0.001, 0.01, 0.1, 1, 10, 100\}$ on official validation by ROC AUC, then average precision, then C closest to 1 on the log scale. The queue-selected model used validation $K = 20$ one-to-one role-qualified recall, followed by many-to-one recall, precision, and the same log-scale tie-break. This procedure selected $C = 10$. Each model was refitted on official train plus validation. Lexicographically ascending candidate ID resolved score ties.

The baselines were: source ResNet weed score; geometry only; role-plus-local specialist features; frozen DINOv2 on the tight RGB bounding box; and component-masked frozen DINOv2. We explicitly distinguish eligible-only fitting from all-candidate fitting. Eligible-only models learn weed versus crop from candidates already satisfying the role contract. All-candidate models learn role-qualified weed versus every other candidate, including crop, mixed-role, background-dominated, and fragmented regions. The latter objective matches the support encountered by the queue.

DINOv2 used the public ViT-S/14 LVD-142M weights distributed through timm, with 224×224 inputs and a frozen encoder [18]. The unmasked representation used the tight candidate bounding box without padding. The masked representation retained proposal-component pixels and filled its surroundings with the per-image RGB median before resizing. Both representations used RGB and proposal geometry exclusively. This paired context ablation measures how surrounding field texture contributes to ranking.

The equal-target-mask-access baseline was a three-class ResNet-18 U-Net initialized from ImageNet-1K and trained on all 104 official-training RGB/mask pairs. Random 512×512 crops and horizontal/vertical flips were applied, with RGB normalization matched to the source models. Its loss combined weighted cross-entropy with weights (0.15, 1, 1) and crop/weed soft Dice. AdamW optimized the model for 40 epochs (learning rate 3×10^{-4} , weight decay 10^{-4}). Official validation selected the epoch, threshold, and component area. Selection first maximized $K = 20$ one-to-one role-qualified recall among settings with mean burden at most 100 candidates/image. Subsequent tie-breaks used many-to-one recall, precision, lower burden, higher threshold, larger area, and earlier epoch.

Training variability was characterized with five independently initialized, 40-epoch single-device runs identified by numeric seeds 20260810–20260814; these identifiers are not acquisition dates. Each run used the same architecture, loss, optimizer, global batch size of six, candidate grid, and validation-only selection rule. The series ran concurrently on two NVIDIA H200 accelerators, with every seed confined to one accelerator and a local batch size of six. Training seed was the independent unit for the stability summary; we report the mean, sample standard deviation, minimum, maximum, and every seed value without treating candidates or instances as independent repeats.

The small-data adaptation recipe trained the final transformer block, final normalization, and binary head from each prospective draw of 300 simulated training-candidate labels derived from dense masks. Model selection additionally used all 2,436 labelled validation candidates; the experiment therefore evaluates a restricted training-label draw under a fully labelled validation set, not a 300-label end-to-end annotation regime. A deterministic rule assigned a positive label only when the candidate satisfied the role-qualified weed contract; all crop, mixed-role, background-dominated, and fragmented candidates were negative. Training used 8 epochs, batch size 64, AdamW, weight decay 10^{-4} , and class-weighted binary cross-entropy. Learning rates were 10^{-5} for the transformer block/norm and 10^{-3} for the head. Random horizontal and vertical flips supplied augmentation. Validation selected maximum candidate-level role-qualified AUC, then average precision, then the earlier epoch. Ten paired repeats used identical selected IDs for frozen and fine-tuned comparisons. The resulting empirical draw distribution characterizes this archived last-block adaptation recipe; it is not interpreted as a stable 95% sampling interval.

2.5. Queue metrics, controls, and uncertainty

For each test image, the first $K \in \{1, 5, 10, 20, 50\}$ candidates formed the fixed queue. Images with fewer than K proposals contributed every available candidate. At the same seed-specific total as fixed $K = 20$, a post-test quota grid crossed minimum allocations $\{0, 5, 10, 20\}$ with maxima $\{20, 30, 50, \text{unrestricted}\}$. The 5–50 policy is retained as an exploratory pooled-utility setting rather than a primary policy. A validation-only rule first required candidate-level role-qualified precision at least as high as validation fixed $K = 20$, then maximized validation worst-date one-to-one recall, pooled recall, and precision in that order. A date-balanced comparator preserved each date’s fixed-queue candidate count and globally ranked candidates within date. Unconstrained global top- N and a validation-transferred score threshold provide additional diagnostics. These policies use scores, tile identities, and acquisition groups, so we report per-tile allocation, date shares, pooled/date-macro/worst-date metrics, and the two released capture-index groups as spatial patch proxies.

Three controls locate the fixed- K ceiling. A 1,000-draw random ranker establishes an ordering reference. An exact integer program maximizes role-qualified recalled instances at each K under one selection decision per candidate. For the crop-constrained oracle, scene frequency α becomes the integer allowance $\lfloor \alpha n_{\text{scene}} \rfloor$. Thus, $\alpha = 0.01$ with 26 scenes permits zero crop-exposed scenes and removes every candidate with positive crop fraction. At $K = 20$, we decompose the distance from perfect recall into proposal loss, budget loss, and the ranking gap to the exact oracle. Global allocation has a broader feasible set because it can assign more than 20 candidates to a tile.

Full-supervision rankers use every official train/validation candidate label derived from dense masks. The equal-target-mask-access semantic baseline trains directly on all 104 target-training masks, with validation choosing epoch, foreground threshold, and minimum component area. Separately, ten simulations of 300 training-candidate labels compare prospective uniform sampling, prospective greedy farthest-first DINOv2 coreset selection, and retrospective class-stratified sampling. Uniform and coreset designs reveal training labels after ID selection, whereas the stratified reference uses class labels during selection; all three use the fully labelled 2,436-candidate validation set. The supervision ledger distinguishes simulated candidate-ranker labels, source masks, target masks, and public pretraining.

The 600×600 tile is the measured experimental unit. The 26 test tiles represent two spatial patches in one field with repeated acquisition positions.

The public release provides capture indices rather than authoritative field coordinates. We therefore summarize pooled, per-date, per-tile, and two-group capture-index proxy results. Date-macro summaries weight the four acquisition dates equally; worst-date values are the minimum for benefit metrics and the maximum for crop exposure. Paired tile differences include the median, interquartile range, full range, and improved/unchanged/worsened counts. Candidate-level role-qualified AUC and average precision are reported by date and by tiles containing both classes. No population-level p -values are calculated from the two patch proxies. Ten-repeat ranges are empirical quantiles across simulated label draws, with paired differences computed within each draw. Group-wise robustness is descriptive because two patches from one field cannot estimate between-field uncertainty [30].

Target-semantic candidate-instance intersections were replayed from the saved binary connected-component masks and official instance rasters. The replay regenerated all 1,347 candidate identifiers and reproduced all 26 fixed-queue scene summaries exactly. Quota and patch-proxy results use these direct mask intersections; bounding-box reconstruction is not used for the reported target-semantic matching results.

3. Results

3.1. *Commissioning increased proposal coverage and candidate burden*

On the official test split, the pre-existing setting proposed 802 candidates (30.85/image), whereas the commissioned setting proposed 2,351 (90.42/image). Spatial weed proposal recall increased from 509/1,712 (0.2973) to 978/1,712 (0.5713). Role-qualified recall increased from 330/1,712 (0.1928) to 518/1,712 (0.3026). Ambiguous or background candidates increased from 0.2506 to 0.7554 of the pool. Role-qualified weed candidates consequently decreased from 0.5648 to 0.1974 of the pool (Figure 2). The coverage gain therefore carried a 2.93-fold increase in candidate burden; Table 2 reports the corresponding fixed-ranker queue comparison.

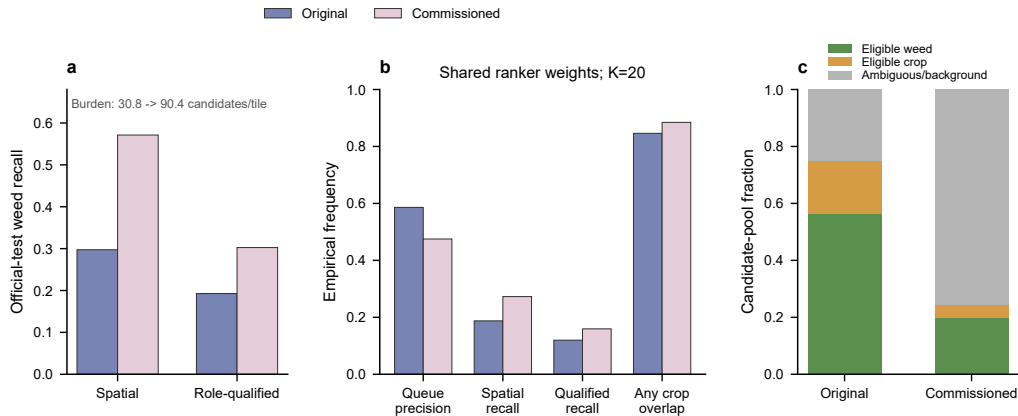


Figure 2: Proposal commissioning on the official test split. (a) Spatial and role-qualified proposal recall with candidate burden. (b) End-to-end comparison at $K = 20$ using identical all-candidate role-plus-local weights. (c) Candidate-pool composition. Crop exposure is the frequency of images containing any queued candidate with nonzero crop overlap.

Table 2: Shared-ranker-weight original versus commissioned queues at $K = 20$. The comparison holds ranker weights fixed but not workload: seven original-setting images contained fewer than 20 proposals, so 471 rather than 520 candidates were accepted.

Setting	Accepted	Weed cand.	Precision	Spatial recall	Qualified recall	Crop scenes
Original ($\tau = 0.95$, 32 px)	471	276	0.5860	321/1712 (0.1875)	205/1712 (0.1197)	22/26 (0.8462)
Commissioned ($\tau = 0.70$, 32 px)	520	247	0.4750	467/1712 (0.2728)	273/1712 (0.1595)	23/26 (0.8846)

3.2. All-candidate training corrected a support mismatch

An eligible-only role-plus-local model achieved high discrimination inside the eligible subset but candidate-level role-qualified AUC 0.4420 and average precision 0.1640 when applied to all 2,351 test candidates. Training the same feature family on the all-candidate weed-versus-rest target increased these values to 0.9049/0.6960. Geometry alone reached 0.8154/0.5418. Thus, the principal transparent-baseline gain came from matching the fitting target to queue support.

Frozen DINOv2 further increased candidate-level role-qualified AUC/AP to 0.9630/0.8684. Component masking reached 0.9665/0.8813. The masked improvement is small but directionally consistent for both metrics, indicating that the discriminative signal remained when context outside the proposal

component was replaced. At $K = 20$, the unmasked and masked rankers achieved role-qualified recall 305/1,712 (0.1782) and 319/1,712 (0.1863), respectively (Table 3; Figure 3).

Table 3: Official-test all-candidate ranking and $K = 20$ queue results for the discrimination-selected rankers. AUC, AP, and precision use the candidate-level role-qualified weed label; selection used official validation only.

Ranker	AUC	AP	Precision	Spatial recall	Qualified recall
Source ResNet score	0.4947	0.2877	187/520 (0.3596)	318/1712 (0.1857)	192/1712 (0.1121)
All-candidate geometry	0.8154	0.5418	212/520 (0.4077)	534/1712 (0.3119)	287/1712 (0.1676)
All-candidate role + local	0.9049	0.6960	247/520 (0.4750)	467/1712 (0.2728)	273/1712 (0.1595)
Frozen DINOv2	0.9630	0.8684	276/520 (0.5308)	431/1712 (0.2518)	305/1712 (0.1782)
Component-masked DINOv2	0.9665	0.8813	278/520 (0.5346)	459/1712 (0.2681)	319/1712 (0.1863)

3.3. Proposal, budget, and ranking each constrained queued recall

At $K = 20$, the role-qualified proposal ceiling was 518/1,712 (0.3026), and exact per-image optimization covered 426 instances (0.2488). The 0.01 crop-scene rule permits $\lfloor 0.01 \times 26 \rfloor = 0$ exposed scenes. Crop-excluding optimization retained 418/1,712 instances (0.2442). The discrimination-selected component-masked ranker recovered 319/1,712 (0.1863), producing a 0.0625 ranking gap to the fixed- K oracle. Relative to perfect recall, proposal construction contributed 0.6974 and the per-image budget contributed 0.0537. Queue-metric selection recovered 324/1,712 (0.1893) and narrowed the ranking gap to 0.0596.

The component-masked ranker exceeded the 1,000-draw random 95% empirical range at $K = 20$ (0.077–0.104 role-qualified recall). Many-to-one recall of 0.1863 became 230/1,712 (0.1343) under one-to-one matching, quantifying candidate–instance multiplicity. Under IoU-only thresholds 0.25/0.50/0.75, many-to-one recall was 0.1898/0.0952/0.0140. Corresponding one-to-one recall was 0.1793/0.0952/0.0140. Across the evaluated role-contract grid, masked-DINOv2 many-to-one recall ranged from 0.0485 to 0.2886. One-to-one recall ranged from 0.0374 to 0.1881. These sensitivity curves position the 0.50/0.50/0.90 contract within the broader operating space.

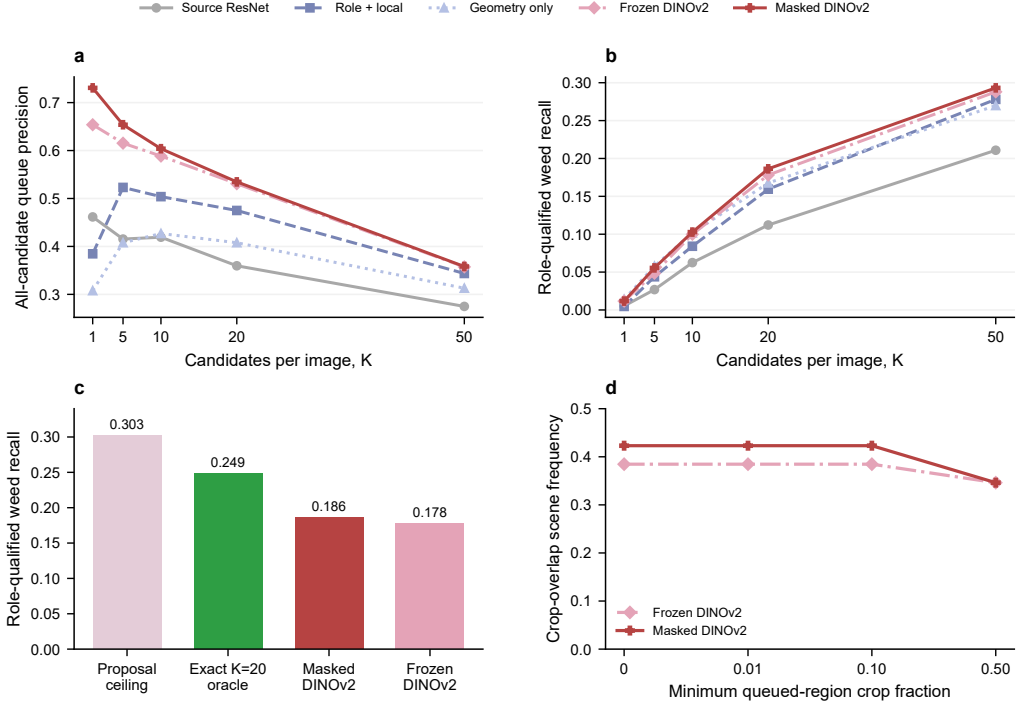


Figure 3: Official-test queue behavior. (a) All-candidate precision and (b) role-qualified recall across K . (c) Proposal, exact-budget, and observed $K = 20$ ceilings. (d) Test-scene frequency with queued crop overlap at four minimum crop fractions.

3.4. Frozen representations remained competitive with 300 training labels and a fully labelled validation set

Across ten official-training draws of 300 simulated mask-derived training-candidate labels, prospective uniform sampling achieved mean candidate-level role-qualified AUC/AP 0.9549/0.8544, $K = 20$ precision 0.5242, and role-qualified recall 0.1724. DINOv2 coreset selection achieved 0.9543/0.8534, 0.5217, and 0.1717. The retrospective class-stratified reference achieved 0.9547/0.8502, 0.5231, and 0.1725. In this candidate pool, the prospective policies produced similar empirical means to the retrospective reference; the ten draws do not establish population-level equivalence (Figure 4).

Under the predefined last-block recipe, validation selected epoch 8 in all ten paired repeats. Fine-tuned minus frozen candidate-level role-qualified AUC averaged -0.0396 , with empirical 2.5/97.5 draw percentiles of $-0.0525/-0.0170$. Precision averaged -0.0373 , and role-qualified recall averaged -0.0115 ,

with reductions in 9/10 repeats. These are paired descriptive summaries of the archived optimizer, trainable block, augmentation, and epoch grid, not confidence intervals.

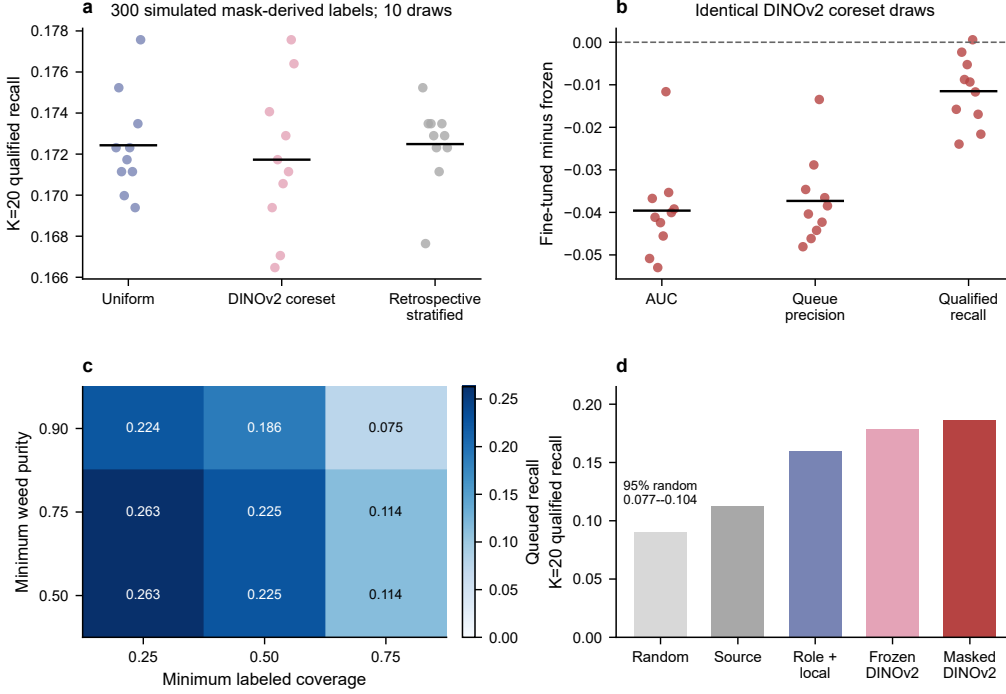


Figure 4: Controls and sensitivity. (a) Ten draws of 300 simulated mask-derived training-candidate labels; every model also uses the fully labelled validation set, and horizontal bars are empirical means. (b) Paired last-block fine-tuning minus frozen-feature differences. (c) One slice of the evaluated matching-contract grid. (d) $K = 20$ role-qualified recall against random ranking and progressively stronger rankers.

Per-date behavior was heterogeneous. For the discrimination-selected masked DINOv2 ranker at $K = 20$, role-qualified recall was 12/392 (0.0306), 84/440 (0.1909), 210/726 (0.2893), and 13/154 (0.0844) across the four acquisition dates. Corresponding precision was 0.0813, 0.5563, 0.9375, and 0.6500. Candidate discrimination also varied: the pooled AUC/AP of 0.9605/0.8617 for the queue-selected ranker became date-macro values of 0.8774/0.6901, with worst-date values of 0.7382/0.2131. Twenty-four of 26 tiles contained both candidate classes and supported tile-level AUC/AP; their macro values were 0.9091/0.6869. These descriptive strata reveal variation hidden by pooled

estimates.

3.5. Quota-constrained allocation traded pooled recovery against temporal coverage

Selecting linear regularization by validation $K = 20$ one-to-one recall chose $C = 10$. Its test candidate-level role-qualified AUC/AP were 0.9605/0.8617, compared with 0.9665/0.8813 under discrimination-based selection. Under fixed $K = 20$, the queue-selected model accepted 520 candidates. Precision was 0.5288, many-to-one qualified recall was 0.1893, and one-to-one qualified recall was 0.1326. The exploratory 5–50 policy reallocated the same 520 slots and increased these pooled values to 0.7346, 0.2646, and 0.1840, respectively (Table 4). Date-macro precision, many-to-one recall, and one-to-one recall changed from 0.5516/0.1504/0.1111 under fixed allocation to 0.6133/0.1959/0.1415 under 5–50 allocation. However, worst-date many-to-one and one-to-one recall decreased from 0.0255 to 0.0153. Unconstrained global top-520 produced pooled values of 0.7462/0.2710/0.1869 and is retained as a diagnostic upper bound. The validation-transferred threshold accepted 426 candidates and achieved precision 0.8169, with many-to-one/one-to-one recall of 0.2477/0.1694.

Table 4: Queue-allocation analysis using the queue-selected masked-DINOv2 ranker. The first three policies share total budget $N = 520$; the validation threshold realized $N = 426$. Macro and worst values summarize four acquisition dates.

Policy	Accepted	Precision	Qualified recall	1:1 recall
Fixed $K = 20$ /tile	520	0.5288	0.1893	0.1326
Exploratory 5–50/tile	520	0.7346	0.2646	0.1840
Diagnostic global top- N	520	0.7462	0.2710	0.1869
Validation score threshold	426	0.8169	0.2477	0.1694

Policy	Date-macro 1:1	Worst-date 1:1
Fixed $K = 20$ /tile	0.1111	0.0255
Exploratory 5–50/tile	0.1415	0.0153
Diagnostic global top- N	0.1435	0.0153
Validation score threshold	0.1311	0.0128

Allocation counts show how the pooled gain formed (Figure 5). The bounded policy assigned 51/149/285/35 candidates to 25 May, 30 May, 6

June, and 15 June; 54.8% of the budget therefore went to 6 June. It assigned at least five candidates to every tile, although eight tiles received fewer than ten, and its allocation Gini coefficient was 0.3817. Unconstrained global top-520 assigned 38/153/293/36 candidates by date, left one tile empty, assigned fewer than five candidates to five tiles, and had Gini 0.4220. For global minus fixed one-to-one recall, 9/26 tiles improved, 12 were unchanged, and 5 declined. Date-level one-to-one recall fell from 0.0255 to 0.0153 on 25 May, increased from 0.1614 to 0.1773 on 30 May and from 0.1860 to 0.3099 on 6 June, and remained 0.0714 on 15 June. Bounded allocation followed the same pattern with a slightly smaller pooled gain. Thus, neither globally scored policy improved the worst acquisition date.

Crop severity changed alongside queue allocation. Fixed $K = 20$ covered at least half of 32/256 crop instances (0.1250), compared with 12/256 (0.0469) for the exploratory 5–50 policy and 10/256 (0.0391) for unconstrained global top- N . The validation threshold covered 5/256 (0.0195). Maximum candidate crop fraction remained 0.919–0.973 across policies, revealing rare severe overlaps beyond scene frequency.

3.6. Target-semantic components and the representation–allocation comparison

Validation selected epoch 36, semantic foreground threshold 0.30, and minimum component area 16 pixels for the archived target-semantic reference run. On official test, background/crop/weed IoU was 0.9778/0.6957/0.7131, yielding three-class mIoU 0.7955 and plant-role mIoU 0.7044. The model produced 1,347 components (51.81/tile). At fixed $K = 20$, 518 candidates were accepted; 355 were role-qualified weed candidates, giving precision 0.6853. Spatial, many-to-one qualified, and one-to-one qualified recovery were 520/1,712 (0.3037), 453/1,712 (0.2646), and 348/1,712 (0.2033). Direct component-mask replay regenerated every candidate identifier and reproduced these scene-level results exactly.

Applying the exploratory 5–50 candidates/tile rule to the target-semantic scores completed the representation–allocation comparison (Table 5). Under its fixed total of 518 reviews, 5–50 allocation increased candidate precision from 0.6853 to 0.8205 and one-to-one recall from 0.2033 to 0.2477. Crop-exposed scenes decreased from 6/26 to 1/26. Date-macro one-to-one recall increased from 0.1891 to 0.2070, whereas worst-date recall decreased from 0.1169 to 0.0510. Representation and allocation therefore contributed separately, and the pooled improvement did not imply uniform temporal improvement.

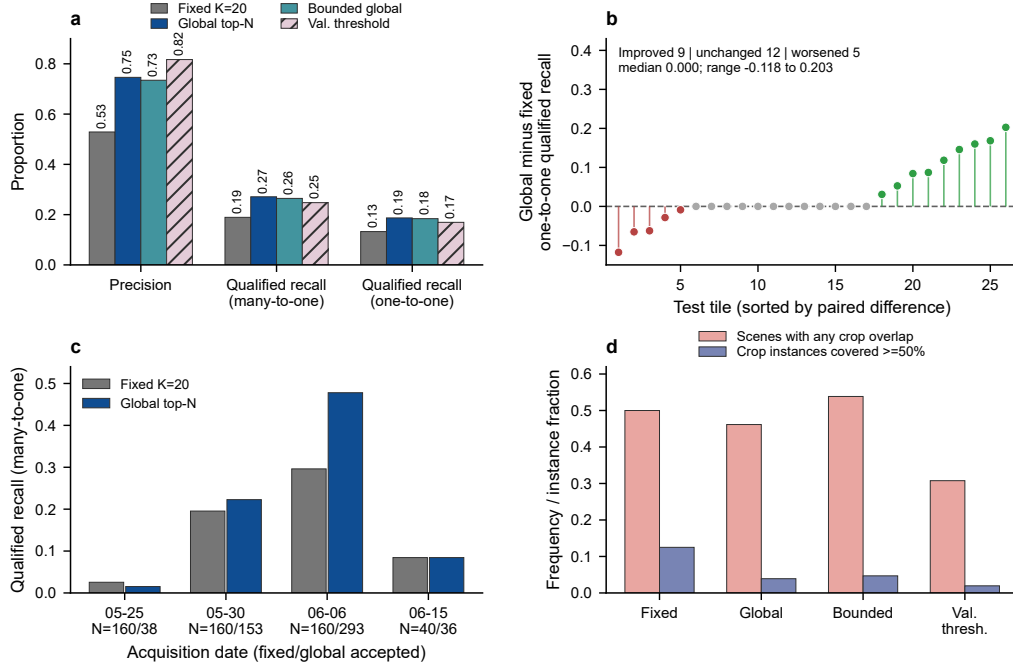


Figure 5: Allocation-policy analysis after selecting regularization by the validation $K = 20$ one-to-one metric. (a) Pooled precision and qualified recall; the validation threshold realizes 426 candidates, while the other policies use 520. (b) Paired tile-level global-minus-fixed one-to-one recall. (c) Per-date many-to-one qualified recall with fixed/global accepted counts. (d) Crop exposure as scene frequency and crop-instance coverage. Unconstrained global allocation and the post-test 5–50 candidates/tile allocation are exploratory diagnostics.

Table 5: Representation–allocation comparison under equal target-mask access. Fixed policies use $K = 20$ /tile; exploratory 5–50 policies preserve each system’s fixed total. Precision counts candidate-level role-qualified weed candidates among all accepted candidates. Recall uses 1,712 weed instances, and crop exposure uses 26 test tiles.

Representation	Allocation	N	Prec.	1:1	Crop	Worst 1:1
Masked DINO	Fixed 20/tile	520	0.5288	0.1326	13/26	0.0255
Masked DINO	Exploratory 5–50	520	0.7346	0.1840	14/26	0.0153
Target semantic reference	Fixed 20/tile	518	0.6853	0.2033	6/26	0.1169
Target semantic reference	Exploratory 5–50	518	0.8205	0.2477	1/26	0.0510

Table 6: Target-semantic H200 stability across five numeric training-seed identifiers. Values are mean $\pm$ sample standard deviation across seeds. Crop is the fraction of 26 test scenes with any queued crop overlap.

Allocation	Accepted	Precision	1:1 recall	Crop	Worst-date 1:1
Fixed 20/tile	511.8 ± 5.6	0.7168 ± 0.0260	0.2027 ± 0.0057	0.2462 ± 0.0439	0.1100 ± 0.0106
Exploratory 5–50/tile	511.8 ± 5.6	0.8939 ± 0.0249	0.2578 ± 0.0090	0.0615 ± 0.0439	0.0473 ± 0.0089
Date-balanced global	511.8 ± 5.6	0.7710 ± 0.0226	0.2192 ± 0.0064	0.2615 ± 0.0834	0.1163 ± 0.0190
Global top- N	511.8 ± 5.6	0.9158 ± 0.0255	0.2643 ± 0.0083	0.0077 ± 0.0172	0.0210 ± 0.0092
Validation threshold	342.0 ± 17.4	0.9487 ± 0.0194	0.1853 ± 0.0137	0.0000 ± 0.0000	0.0077 ± 0.0065

3.7. H200 repetitions exposed a quota-dependent pooled-group trade-off

Five independent H200 runs selected epochs 31–40; the 16-pixel component minimum was selected in all runs, while the foreground threshold was 0.30 once, 0.50 three times, and 0.70 once. Official-test semantic mIoU was 0.7955 ± 0.0127 (mean $\pm$ sample SD). At fixed $K = 20$, candidate-level role-qualified precision was 0.7168 ± 0.0260 and one-to-one recall was 0.2027 ± 0.0057 (Table 6).

Across the full 4×4 quota grid, increasing the maximum allocation increased pooled recovery but generally weakened the lowest-recall date (Figure 6). The 5–50 setting improved precision by 0.1770 ± 0.0146 and one-to-one recall by 0.0551 ± 0.0053 relative to fixed $K = 20$, with positive differences in every seed and both patch proxies. It reduced worst-date one-to-one recall by 0.0627 ± 0.0070 in every seed. Date-balanced allocation provided a less concentrated alternative: precision 0.7710 ± 0.0226 , one-to-one recall 0.2192 ± 0.0064 , and worst-date recall 0.1163 ± 0.0190 . Relative to fixed allocation, its one-to-one recall effect averaged -0.0458 in patch proxy 0 and $+0.0603$ in patch proxy 1. Validation selection chose fixed $K = 20$ in four seeds and a 0–30 quota in one; it never selected 5–50. These results identify the 5–50 gain as an exploratory pooled-utility trade-off rather than a validation-supported group-robust policy.

The official WeedsGalore paper reports RGB semantic mIoU of 79.33% for DeepLabv3+ and 79.26% for MaskFormer, with RGB MaskFormer instance mAP of 32.17% (AP50 33.66%, AP75 32.01%) [14]. Table 7 places the archived target-semantic reference beside those published values. Because semantic mIoU, instance mAP, and role-qualified queue recall characterize non-interchangeable evaluation levels, the table presents them as complementary context.

Table 7: Published official WeedsGalore RGB context and the target-semantic baseline.

System and evidence tier	Metric	Value (%)
Official DeepLabv3+ RGB (published)	Semantic mIoU	79.33
Official MaskFormer RGB (published)	Semantic mIoU	79.26
Official MaskFormer RGB (published)	Instance mAP	32.17
ResNet-18 U-Net RGB (single-run target-semantic baseline)	Semantic mIoU	79.55
ResNet-18 U-Net RGB (single-run target-semantic baseline)	Queue 1:1 recall at $K = 20$	20.33

4. Discussion

The official spatial analysis identifies distinct transitions from proposals to finite review queues. Commissioning approximately doubled spatial recall, while candidate burden tripled and three quarters of the pool became ambiguous or background-dominated. Support-aware all-candidate fitting aligned supervision with the deployed queue and increased role-plus-local candidate-level role-qualified AUC from 0.4420 to 0.9049. Candidate support is therefore a central design variable alongside representation quality.

The ceiling decomposition turns aggregate recall into an engineering diagnostic. Missed proposals form the largest distance from perfect recovery. Within the commissioned pool, the ranking gap slightly exceeds the remaining $K = 20$ budget gap. Component formation is therefore the primary improvement target, paired with ranker fitting on the complete candidate distribution. The shared-ranker-weight comparison further quantifies how proposal density trades coverage against crop exposure and workload; its accepted workloads differ because seven original-setting tiles supplied fewer than 20 candidates.

Queue allocation emerged as a separate optimization problem. With candidate scores and total review budget fixed, globally scored policies shifted capacity toward the high-density 6 June acquisition. The complete quota grid shows a continuous pooled-recovery versus weakest-date frontier. The exploratory 5–50 policy increased pooled precision and one-to-one recovery

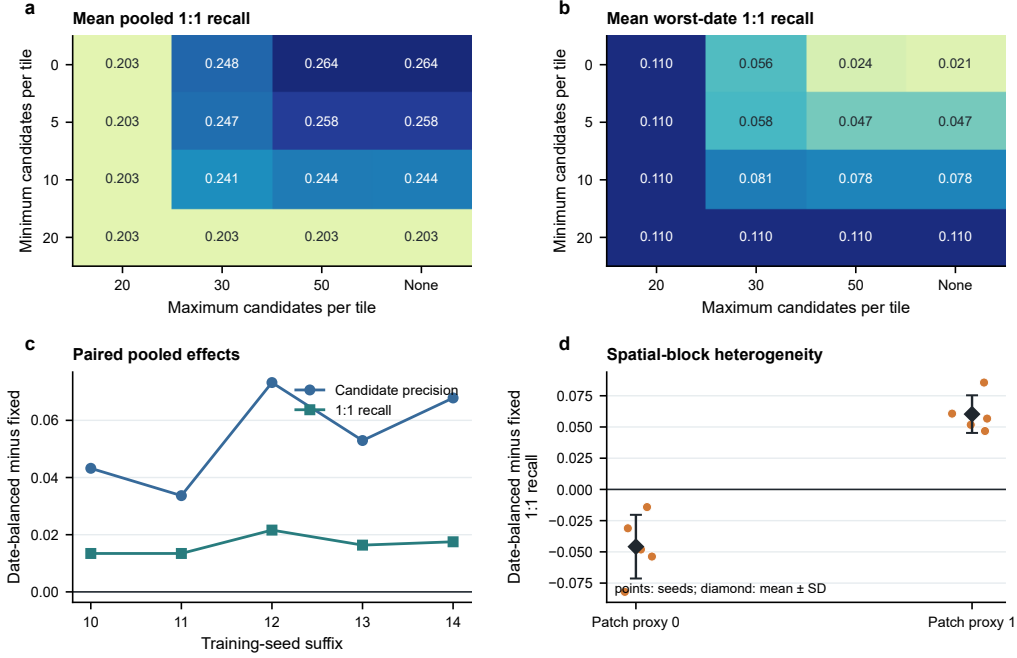


Figure 6: Exploratory target-semantic quota and group robustness across five H200 training seeds. (a,b) Mean pooled and worst-date one-to-one recall over the complete minimum/maximum quota grid. (c) Paired date-balanced-minus-fixed effects for candidate-level role-qualified precision and one-to-one recall. (d) Date-balanced-minus-fixed one-to-one recall in the two capture-index patch proxies; points are seeds and diamonds with error bars show mean $\pm$ sample SD.

in every H200 seed and in both patch proxies, while consistently reducing worst-date recovery. Date-balanced allocation preserved mean worst-date performance and modestly improved pooled metrics, but its effect reversed between the two patch proxies. Validation selection consequently retained fixed $K = 20$ in four of five seeds. These findings align with group-robust learning results in which pooled optimization and worst-group performance represent distinct objectives [30].

Direct target-semantic training changed the preferred component representation. The archived reference recovered more one-to-one weed instances than the source-specialist proposal plus masked-DINO ranker, and the five H200 repetitions placed fixed-queue recall in a narrow 0.1951–0.2097 range. Exact component-mask replay produced identical candidate identifiers and

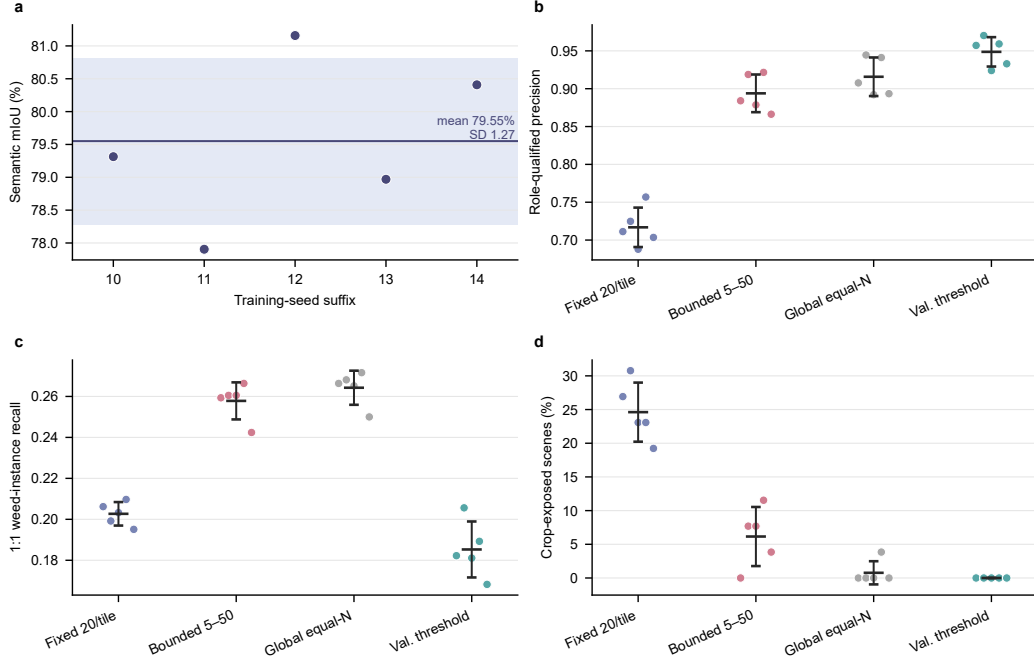


Figure 7: Five-seed H200 stability of target-semantic component formation and queue allocation. (a) Official-test semantic mIoU for each numeric training-seed identifier. (b–d) Candidate-level role-qualified precision, one-to-one weed-instance recall, and crop-exposed scene frequency for fixed, exploratory 5–50, global, and validation-threshold policies. Points denote independent training seeds; bars show the mean and sample standard deviation.

scene metrics, removing geometric reconstruction from this comparison. The source-specialist representation remains a single recorded-run comparator, whereas the target-semantic series quantifies initialization variability.

Component masking changed DINOv2 AUC/AP by 0.0035/0.0129 in the recorded run. This paired result functions as a context ablation. The broader contribution is the measurable separation of spatial discovery, role qualification, rank ordering, and budget allocation. The same separation clarifies supervision cost: proposal-setting selection uses complete official-training masks; full rankers use 9,887 training and 2,436 validation candidate labels; the nominal 300-label experiments restrict only the sampled training candidates and retain the fully labelled validation set. This supervision ledger positions the queue analysis alongside methods that reduce label construction at the segmentation-training stage [21, 22, 23].

The evidence supports a multitemporal single-field engineering case study.

Four dates characterize temporal variation within one maize field, and two capture-index groups provide spatial-block diagnostics within the official test split. The five training seeds quantify optimization variability rather than between-field uncertainty. Independent farms, seasons, crops, and sensors define the next validation level. Within the present scope, timestamped configurations, exact mask replay, failure records, source-data tables, and group-stratified outputs make the computational chain auditable.

5. Conclusion

This multitemporal single-field case study connects candidate formation, role-qualified ranking, and review-budget allocation within one crop–weed queue. Proposal commissioning increased recoverable support at a substantial candidate-burden cost. Across five H200 seeds, the exploratory 5–50 allocation increased pooled target-semantic one-to-one recall from 0.2027 ± 0.0057 to 0.2578 ± 0.0090 and precision from 0.7168 ± 0.0260 to 0.8939 ± 0.0249 , while reducing weakest-date recall. Date-balanced allocation preserved mean weakest-date performance but produced opposite recall effects in the two spatial patch proxies. Support-aware fitting, exact component matching, and group-stratified allocation therefore provide complementary controls for interpreting finite crop–weed review queues.

Data availability

SugarBeets2016 and WeedsGalore are public datasets available from their cited official sources. The official SugarBeets page identifies the release as CC BY-SA 4.0, and WeedsGalore includes a CC BY 4.0 licence. The reproducibility archive records source URLs, download dates, release identifiers, licence snapshots, and SHA-256 checksums. Source imagery remains available through the official repositories; derived source-data tables and verification hashes accompany the code release.

Code availability

The versioned reproducibility archive is available from GitHub Release v1.1.0 at <https://github.com/Fytap/agrispec-vlm-reproducibility/releases/tag/v1.1.0>. It contains frozen configurations, environment specifications, training and embedding-extraction source, model acquisition

instructions, archived prediction and exact matching outputs, table/figure scripts, run summaries, failure records, and clean-room replay commands. Public dataset imagery and public model weights remain at their cited upstream sources.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used OpenAI Codex for code review, reproducibility checks, and language editing, including checks of sentence clarity, grammar, and wording. The authors reviewed and edited all affected content and take full responsibility for the content of the publication.

Ethics declaration

This study analysed public agricultural image datasets and did not involve human participants, animals, or personally identifiable information.

Declaration of competing interest

Zhi Ling, Yu Yan, and Ziyi Kuang report an affiliation with Kiwiar Co., Ltd. No proprietary dataset or commercial product was evaluated in this study.

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